Blame the Data, or the Model? Diagnosing a Domain Corpus's Syntax Before Training

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Abstract

When a language model fails to generalize a pattern of its domain, the cause can lie in the model or in the corpus, and existing data-quality measures cannot tell these apart. A knowledge domain, such as recipes, is a bounded subset of a language's patterns; we address its syntactic layer. We propose a corpus-only indicator of local syntactic sufficiency. It reads every local syntactic pattern of a corpus at two levels, a head followed by a grammatical class (*she* → verb) and a class of heads induced from the corpus (pronoun → verb), and compares the variety of its fillers with size-matched random samples of a reference corpus; without a reference, each head is compared with its siblings in the corpus. No threshold is set by hand: the tokenizer's granularity, the least evidence read and the number of random samples are derived from the data, leaving only the test's level. We validate the indicator with preregistered corpus interventions and trained models. On a synthetic grammar it predicts which patterns a model generalizes (Spearman 0.94–0.98), while held-out loss ranks the corpora in reverse. On FineWeb it finds planted gaps without false alarms, at heads where models lose 1–4.6 nats on the true next word. Removing a pattern from a whole class removes it from every member, while removing it from two heads costs them only 25–57% as much: local syntax is learned mostly at the class level. When the indicator flags a class pattern, no model configuration we tried recovers it; when it flags nothing, a model reaches the reference's syntax within 0.02. Unlike one-number corpus metrics, the indicator localizes the damaged pattern. Its scope is adjacency syntax, validated with models of up to 6.2M parameters.

Keywords: knowledge domains; domain-specific corpora; syntactic sufficiency; training data diagnostics; syntactic generalization; filler diversity; language models

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1. Introduction

A practitioner who trains a language model on a domain corpus and finds that it fails to generalize faces a diagnostic question before any engineering one. Is the corpus missing something, or is the model wrong for it? The two call for opposite actions. Collecting or rebalancing data cannot fix a model that is too small or badly trained, and tuning the architecture cannot fix a corpus that never shows a pattern with enough variety. Today this question is answered, if at all, by training and inspecting.

We frame the question through the notion of a **knowledge domain**: a bounded subset of the syntactic and semantic patterns of a language, such as recipes, which are mostly imperative actions on ingredients. This paper addresses only the syntactic layer of a domain. By syntax we mean the well-formed frame of a sentence regardless of its truth. "Cut the milk" is syntactically correct; that it makes no sense is a matter of semantics and lies outside our scope. Our premise follows a long-standing obser-

vation in corpus linguistics (the productivity measures of Baayen, 2009) and in language-model learning (Wei et al., 2021; Misra & Mahowald, 2024): a pattern is learned as a pattern, and not as a memorized word, when the corpus shows its slot with enough different fillers. We call a corpus **syntactically sufficient** for its domain when it shows every local pattern of the domain in this way.

Existing tools do not measure this. Data-quality and selection methods, such as perplexity filtering in the CCNet style (Wenzek et al., 2020), DSIR, QuRating and the diversity coefficient, give one number per document or dataset, aimed at perplexity or downstream accuracy, and a single score cannot say which pattern is missing. Held-out loss can mislead: on a controlled grammar we show that it ranks corpora in reverse of syntactic generalization, because a restricted corpus is restricted in its held-out split too and therefore looks more predictable. Finally, filtered-corpus training (FiCT; Patil et al., 2024) and related studies remove a construction and test what the trained model learns. These studies are post-hoc by design, and their main message, that models generalize from indirect evidence, makes the question of which removals matter a real one.

A diagnostic of this kind is only as trustworthy as its settings. A threshold tuned on the corpora where the method was developed can produce, on a new corpus, flags that are artifacts of the threshold. We therefore require every setting of the indicator either to be derived from the data or to be shown not to change its verdicts, and we report the measurements (§5.8).

Our contributions are the following:

1. **An indicator computed on the corpus alone.** It reads local syntactic patterns at the head level and at every node of an induced tree of heads, comparing them with size-matched random samples of a reference or, without one, with sibling heads in the corpus (§3); it is packaged as a command-line tool and a user interface.
2. **Settings from the data.** A measured range of tokenizer granularity, a test level that sets the number of random samples, and a trunk read chosen by Tarone's criterion replace every hand-set threshold (§3.4, §5.8).
3. **Causal validation by corpus interventions** in a synthetic grammar and in natural text, covering planted gaps, dose–response, realistic filters, and removal of a pattern from a head versus from its whole class (§5).
4. **A finding about the level at which syntax is learned.** A head that loses its slot keeps its syntax while its class keeps the pattern, whereas a class that loses the pattern loses the syntax of every member (§5.4).
5. **An attribution test.** A flagged class pattern is a corpus fault that no tried model configuration fixes, and an unflagged corpus lets the reference's syntax be reached (§5.5); this is the guarantee a practitioner needs.
6. **Comparisons with one-number corpus metrics and BLiMP**, with an honest account of where they do and do not see the same thing (§5.7), together with a list of every failed prediction (Appendix A).

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2. Related work

Data quality and selection for pretraining. Perplexity filtering against a reference language model, as in the CCNet pipeline, is a standard quality filter. DSIR (Xie et al., 2023) selects data by importance resampling on hashed n-gram features, and QuRating (Wettig et al., 2024) uses language-model judgments of writing quality, expertise and educational value. The diversity coefficient (Miranda et al.,

2023) measures a dataset's variability through Task2Vec embeddings. Training-free predictors of data influence, such as DataProphet (Qi et al., 2026), rank supervision datasets before training. Albalak et al. (2024) survey the field. All of these methods score documents or datasets as wholes. None reports which syntactic patterns a corpus under-serves, and none separates a corpus fault from a model fault. We find that perplexity-based filters are the only cuts of a corpus the indicator flags (§5.3).

Learning syntax from manipulated corpora. Wei et al. (2021) show that BERT applies subject–verb agreement to pairs never seen together, but struggles with infrequent items and strong priors. FiCT (Patil et al., 2024) removes a construction from the training corpus and finds that both LSTMs and transformers generalize to it from indirect evidence; Luo & Steinert-Threlkeld (2026) extend this result to unlike coordination. Misra & Mahowald (2024) show that a rare construction (AANN, "a beautiful five days") is learned from related, more frequent ones, and is learned better with more variability in the input. Our results agree with these findings at the head level: a pronoun that loses its verb slot keeps its syntax through its siblings. We add the class level, where removal does prevent learning, and an indicator that says before training which removals matter.

Productivity and diversity measures. Baayen's (2009) productivity measures use hapax legomena to estimate how open a morphological category is. The Good–Turing missing mass (Good, 1953), singletons over trunk, applies the same idea to a syntactic slot; we show that on its own it does not flag a thinned slot (§5.9). Measures of the lexical, syntactic and semantic diversity of model outputs (Guo et al., 2025) assess generated text rather than training corpora, and they are not tied to what a model learns.

Multiple testing with discrete statistics. The indicator tests tens of thousands of patterns at once. Its bar is the maximum statistic over random samples, a resampling family-wise bar (Westfall & Young, 1993), and patterns too rare ever to be flagged are left out of the family, following Tarone (1990).

Targeted syntactic evaluation. Minimal pairs (Linzen et al., 2016; Marvin & Linzen, 2018), the systematic assessment of Hu et al. (2020) and BLiMP (Warstadt et al., 2020) evaluate trained models. Our model-side reading is a minimal-pair score built to exclude semantics: unseen class-valid fillers compete against frequency-matched ungrammatical ones, with classes taken from an external tagger (§4.3). We also report BLiMP.

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3. The indicator

Figure 1 summarizes the indicator; the subsections below define each step.

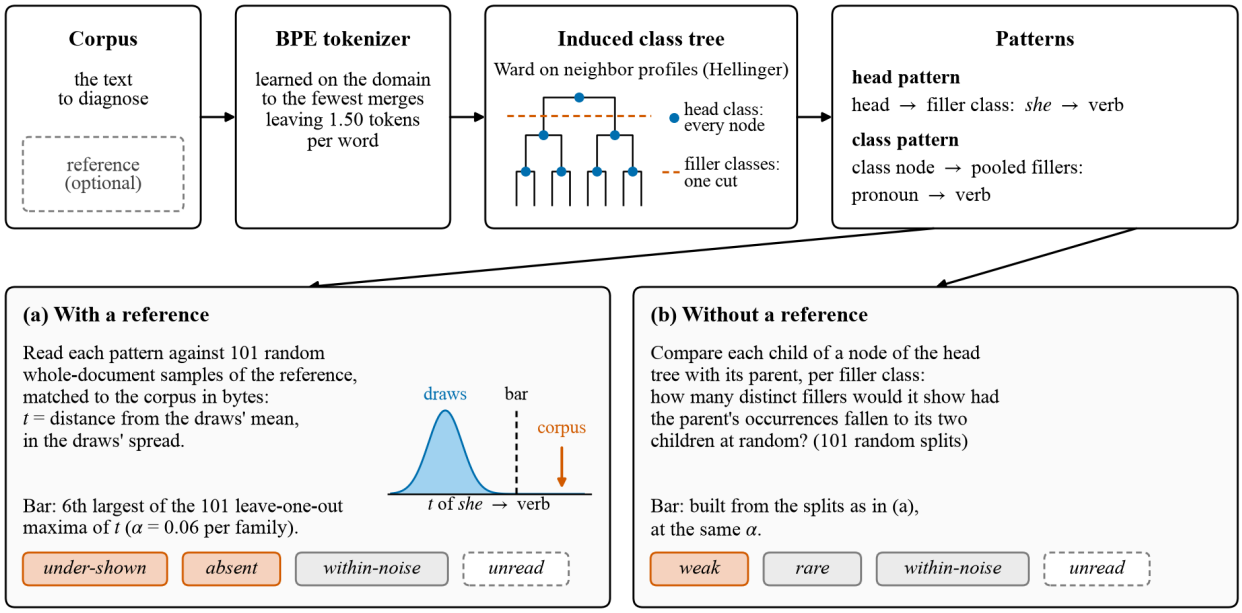


Figure 1. The indicator. The corpus is tokenized at 1.50 tokens per word and its tokens are clustered into one tree: every node is a class of heads, and one cut gives the filler classes. Each head pattern and class pattern is read (a) against 101 size-matched random samples of a reference, with a bar taken from the samples' own leave-one-out maxima at level α, or (b) without a reference, against random splits of the head's parent in the tree. The inset is schematic.

3.1 Units: tokens, transitions, classes

The corpus is tokenized with a byte-level BPE (Sennrich et al., 2016) learned on the domain's text, which is the reference's text or, without a reference, the corpus's own. The tokenizer is trained to the fewest merges that leave 1.50 tokens per word; on the FineWeb pool this is 11,381 merges. We measured that the verdicts do not change from 1.35 to 1.70 tokens per word in five domains, given a domain of at least 14.4 MB (§5.8). Every token takes part in the analysis: any token found in at least two documents can be a head or a filler, whether it is a word, a word piece, punctuation or a number.

A transition is the head's last token followed by the filler. We call it *syntactic* when the filler starts a new pre-token of the tokenizer (a new word, punctuation run, number or contraction), and *lexical* when the filler continues the word of the token before it (`model` → `s`). Because no merge crosses a pre-token boundary, this rule is exact. The two kinds are counted as separate patterns, and the syntactic rows form the diagnosis.

Classes come from a single tree. Each token is described by its left and right neighbor distributions over the most frequent 1,000 tokens. The rows are square-rooted, so that Euclidean distance is Hellinger distance, and are clustered with Ward's (1963) method into one tree. Filler classes are obtained by cutting the tree at the largest jump in merge height between 8 and 64 classes, which gives 12 classes on FineWeb. On the recipe grammar this cut recovers the grammar's categories exactly. A finer cut starts grouping fillers by meaning (what can be poured against what can be chopped), which would bring semantics into a syntactic measure. Head classes, by contrast, require no cut: every internal node of the same tree is a class of heads, and class patterns are read at every node (§3.2).

3.2 Patterns

A head pattern is a head of one or two tokens followed by a filler of class Y, together with its transition kind. Occurrences are counted in distinct contexts, defined by the token before the head and the token after the filler, so that exact copies add nothing. We call a pattern's number of distinct contexts its

trunk. A class pattern is a node of the head tree together with the fillers that follow any of its one-token heads, pooled.

3.3 Readings

The first reading is diversity, defined as fillers / expected fillers, where expected fillers = $\Sigma_y [1 - (1 - p_y)^n]$, with n the trunk and p_y the class shares of the corpus. It compares heads of different traffic against free draws from the class.

The second reading is the divergence $D = KL(R \| S)$, in nats. Here R is the reference's filler distribution for the slot, and S is the corpus's, with Witten–Bell backoff (Witten & Bell, 1991) to the corpus's filler unigram, adding one count per reference filler. D has the units of the loss: a model that learned S and is scored on text from R loses on average $KL(R \| S)$ nats per slot occurrence. On planted gaps the drop in the model's log probability is about $-0.68 \cdot D$ ($R^2 = 0.77$, Appendix B).

Every reading depends on the corpus's size, so a smaller corpus always reads worse. We therefore compare each corpus with size-matched draws: random whole-document samples of the reference of the same size in bytes. The reference is tokenized and classified once and reused for every corpus diagnosed against it.

3.4 Verdicts with a reference

Each pattern's t is how far the corpus falls from the draws' mean, in units of the draws' spread. The reading is diversity for head rows, trunk for *absent*, and D for class rows. The spread includes the reading's own rounding, since diversity moves in steps of one filler and a trunk in steps of one: $sd' = \sqrt{(sd^2 + step^2/12)}$.

The bar comes from the draws. Each draw in turn is left out and read against the others, and its largest t over all patterns is recorded. A corpus is flagged only where it strays further than the k-th largest of these maxima, so that a random sample of the reference passes the bar with probability about $k / (n + 1)$.

The one setting the user states is alpha, the chance that a random sample of the reference is flagged in a family of verdicts. Its default is 0.06 per family, the families being head rows, *absent* and class rows. Alpha is the test's level, a tolerance rather than a property of the data. The number of draws follows from alpha. The k-th largest of n draw maxima is itself noisy: over draw sets, the chance of passing it follows $Beta(k, n + 1 - k)$. The tool uses the fewest draws that keep that chance within alpha/2 to $1.5 \cdot$ alpha for 80% of draw sets, which gives 101 draws and $k = 6$ at alpha 0.06. The trunk read also comes from the draws, by Tarone's criterion. A pattern too rare to be flagged even if the corpus showed a single filler adds no verdict, and its noise raises the bar for every other pattern. The least trunk read is therefore the one that leaves the most patterns that could be flagged.

A head row is *under-shown* when its diversity lies below the draws' mean by more than the bar. It is *absent* when its trunk lies below the draws' mean trunk by more than the *absent* bar: a random sample of this size would show the pattern, and the corpus shows too little of it. Otherwise it is *within-noise*, and below the trunk read it is *unread*.

A class row reads D over the node's pooled fillers against the draws, and a node is *under-shown* when its t exceeds the class bar. Of each branch, the node with the highest t is reported, so that the report names the smallest group of heads that carries the shift. Each class row also lists its *short heads*, the members whose own rows are under-shown or absent.

3.5 Verdicts without a reference

A pattern the corpus shows weakly is weak for a model trained on it, whatever the domain does. A corpus alone can reveal this by comparing a head with its siblings. Each child of a node of the head tree, whether a head or a class of heads, is compared with its parent, per filler class and transition kind. The null asks how many distinct fillers this child would show had the parent's occurrences of that class fallen to its two children at random. To answer it, 101 random splits are drawn exactly, without replacement, at the child's trunk. The bar is built from the splits as in §3.4, at the same alpha, and only tests that a single filler could fail are read.

Two further restrictions apply. Only children of a node inside one filler class are tested: above the class cut, the two sides of a split are different categories (`,/the/to` against `./:`), and every such test read as weak. Heads of two tokens are placed by their first token.

A pattern is *weak* when it lies past the bar, and *rare* when it is shown fewer times than any slot the test can read. Otherwise it is *within-noise*, or *unread* when it has no sibling to read against.

Three cases are invisible without a reference. A pattern the corpus lacks altogether, and a class shown weakly as a whole, leave no stronger sibling to compare with. A head restricted enough for the tree to set it apart from its siblings loses its sibling. The report states these limits and recommends a reference from the same domain. A reference also separates the corpus's own weakness from the domain's: most weak patterns of a natural corpus reflect its grammar (`to` against the other prepositions, reporting verbs), and a reference reads them within noise.

3.6 The tool

`corpus-diagnose` (Python) takes a corpus and, optionally, a reference. It writes JSON and Markdown reports covering a summary, warnings, classes, head rows, class rows and settings, including alpha and the number of draws. When no merge budget is given, it learns the domain's tokenizer to 1.50 tokens per word, resuming the BPE in doubling rounds. A budget given with `--merges N` is checked before the diagnosis: outside 1.35–1.70 tokens per word the command refuses and names the granularity reached, unless `--force` is passed. It warns when the domain, the reference or else the corpus, holds less than 14.4 MB (§5.8). A Streamlit page runs the same diagnosis on any catalogued dataset.

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4. Experimental setup

4.1 Corpora

We use a synthetic recipe grammar, natural text from FineWeb, and a set of interventions on the latter.

The recipe grammar is a probabilistic grammar with a closed lexicon (imperative verbs, countable ingredients, adjectives, adverbs, tools) and four sentence rules. A single knob changes how many ingredients each verb is ever seen with (1, 2, 4, 8, 16, 20, 24 or all), while every ingredient still occurs in other sentences. The reserved verb–ingredient combinations are never seen and serve as test items.

Our natural text is drawn from FineWeb (Penedo et al., 2024). The main pool F consists of 4,600 documents (14.4 MB, ≈2.4M words, 3.4M tokens) from the first 5,000 streamed documents; the other 400 documents are held out for evaluation. A 6.5 MB sample, divided into 8 slices, is used for plantings. BigF is the first 20,000 documents minus the evaluation set (57.8 MB, ≈10M words).

The interventions are of four kinds. In plantings, the followers of chosen heads are replaced by one of their k most frequent fillers, or the sentences in which they take another filler are removed. Document cuts at 10–50% of F are random or made by perplexity filters or length filters. Sentence-level cuts of a

50% cut of F (r50) remove every sentence in which a listed head is immediately followed by a spaCy-tagged class (Honnibal et al., 2020); they are made at the class level, at the head level (two members only) and as random controls, all at matched bytes. Finally, the *labeled cuts* for the tool comprise the class and head cuts of pronouns and modals, possessives, numbers and base verbs (8 cuts), 12 random controls, and two cuts that drop four words each.

4.2 Models

The model is a decoder with the GPT-2 layout (Radford et al., 2019) and a tied head, trained on CPU, in four configurations:

- **small:** 2 layers, 128 wide, 4 heads, context 128, 1.9M parameters, 3,000 steps of 32 × 128 tokens, learning rate 1e-3;
- **medium:** 4 layers, 256 wide, 6.2M parameters, learning rate 5e-4;
- **long:** small, 9,000 steps;
- **scale:** medium, 30,000 steps on BigF, about 5 hours per run.

Each corpus learns its own BPE (11,563 merges on FineWeb), and the best checkpoint is chosen on a held-out fraction. We use seeds 0 and 1 unless noted. On the recipe grammar, whose tokenizer stops at 1,000 merges, the small and medium models have 0.5M and 3.3M parameters.

4.3 Model-side readings

We read trained models in four ways. The *drop* of a head is the mean, over its evaluation occurrences, of log p_model(true next word) − log p_reference model, in nats. It is a filler-level reading, and therefore mixes syntax with semantics and topic.

On the recipe grammar, the *generalization ratio* is the probability the model gives, after "⟨bos⟩ cut the", to ingredients never seen with that verb, relative to their share; 1 means that the pattern is learned and 0 that it is memorized.

On natural text, *G* is read after a real evaluation prefix ending in head h. It is the share of comparisons in which the model prefers a word of h's slot class that never follows h in the reference over a word of an ungrammatical class, matched in log frequency. Truth does not enter: after "she", any past-tense verb is valid. Classes come from spaCy (`en_core_web_sm`), never from the indicator. The grammatical contrasts are:

- pronouns → past-tense verbs (VBD), against nouns and adjectives;
- modals → base verbs (VB), against nouns and adjectives;
- possessives → nouns, against adverbs and VBD/VBZ verbs.

Finally, we score *BLiMP* (67 paradigms × 1,000 pairs) by total log probability.

4.4 Protocol

Every experiment's design, predictions and thresholds were written in the design documents before it ran. Construction parameters were changed only from corpus-side counts, before any model was scored, and each change is recorded. Failed predictions are kept and listed in Appendix A.

The indicator was first built with hand-set thresholds (a diversity floor of 0.5, a 3-sigma margin over 8 draws, a minimum trunk of 20, a class floor of 0.5 nats). Those were replaced by the settings of §3.4–3.5 (§5.8), and every corpus-side experiment was rerun with the final tool. Models were not retrained,

because the corpora are the same and only their reading changed. The earlier version's results are collected in Appendix B; the two corpus-side numbers in §5 that still come from it are marked.

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5. Results

5.1 Held-out measures do not see syntactic insufficiency (recipe grammar)

Measure	Result
Per-verb coverage vs. generalization ratio	Spearman 0.891 (one seed); 0.940–0.975 over 3 seeds × 2 sizes
Mean ratio, all ingredients vs. one per verb	0.909 vs. 0.011
Held-out loss and n-gram predictability	Rank the variants in **reverse** on every seed and size (Spearman 0.98–1.00 between loss and ratio)

The model generalizes late. The small model's ratio at 16 fillers is only 0.17–0.19, and it crosses 0.5 at coverage 0.67–0.80. The larger model needs more, reaching 0.554 against 0.901 at 24 fillers, so that "sufficient" depends on model size, a point we return to in §6. Minimal pairs of the type "cut the tomato / cut the slowly", meant as an easy control, tracked the ratio (0.25 with one filler) instead of staying near 100%. Figure 2 plots both readings.

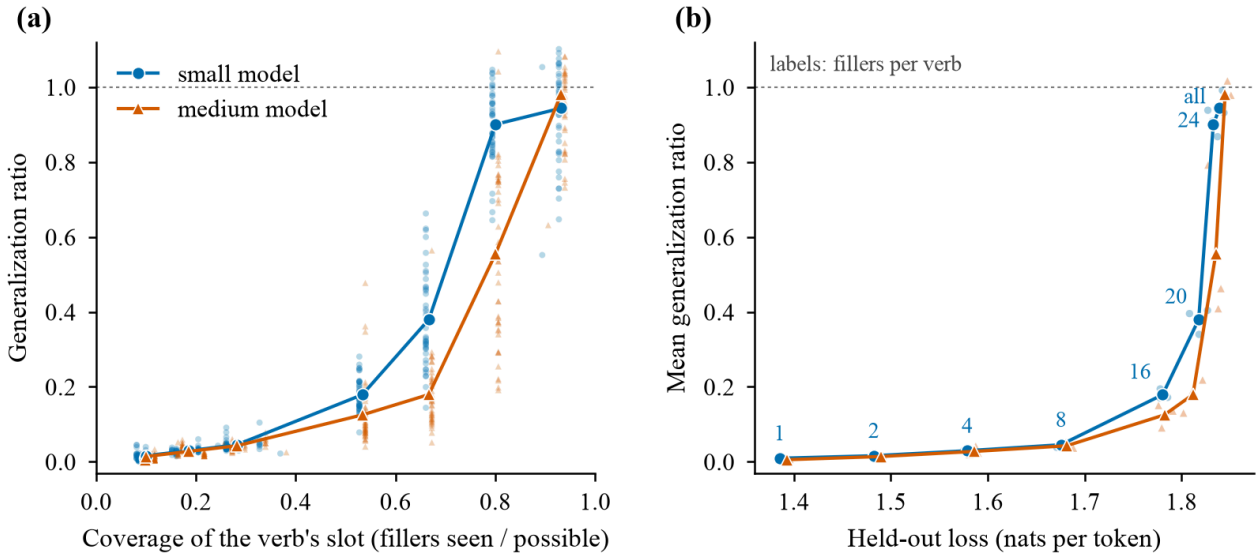


Figure 2. Recipe grammar. (a) Per-verb coverage of the ingredient slot against the generalization ratio, over three seeds and two model sizes (points: verbs; lines: variant means). Coverage predicts which verbs generalize (Spearman 0.94–0.98), and the larger model needs more coverage. (b) Across corpus variants (labels: ingredients per verb), lower held-out loss goes with lower generalization: the most restricted corpora are the most predictable.

5.2 Head level on natural text: planted gaps and dose

Detection, with the final tool. We planted five heads of a FineWeb sample (*people, she, But, only, even*) in one slice and diagnosed each slice against the sample.

Planting	Heads found (of 5)	Other rows flagged
sentences removed, k = 2	4	0
fillers replaced, k = 2, 4, 8, 16	5 at each k	0

Planting	Heads found (of 5)	Other rows flagged
fillers replaced, k = 32	4	0
7 unplanted slices	0	0

Every found head is found through its row of determiner fillers, at diversity 0.03–0.17 for k = 2 and 0.27–0.35 for k = 32, against about 0.9 in the unplanted slices. Most are also read *absent* in a second filler class. No row outside the planted heads is flagged in any slice, of 2,900–6,600 rows read. The class level flags one class row in each of two of the seven unplanted slices (§7).

Model cost. Models trained on the planted corpora give the true next word 2.17–4.60 nats less (k = 2) and 1.12–3.60 nats less (k = 16) at every planted head and seed, while comparison heads stay within 0.3 nats in 19 of 20 readings. The whole-text loss moves only 0.03 bits per byte: it registers the damage, but does not say where.

Dose. We planted fourteen heads to diversity 0.1–0.7, with two assignments and two seeds. The loss is monotone in the dose (Spearman 0.76–0.92), and heads at 0.1 lose 2.8–3.3 nats. The mild end is not free: heads at diversity 0.5–0.7 lose 0.38–0.57 nats; diversity is therefore not the effect size. D is the effect size (drop $\approx -0.68 \cdot D$, R^2 0.77, earlier reading, Appendix B), although on plantings it orders heads no better than diversity. Figure 3 shows every planted head.

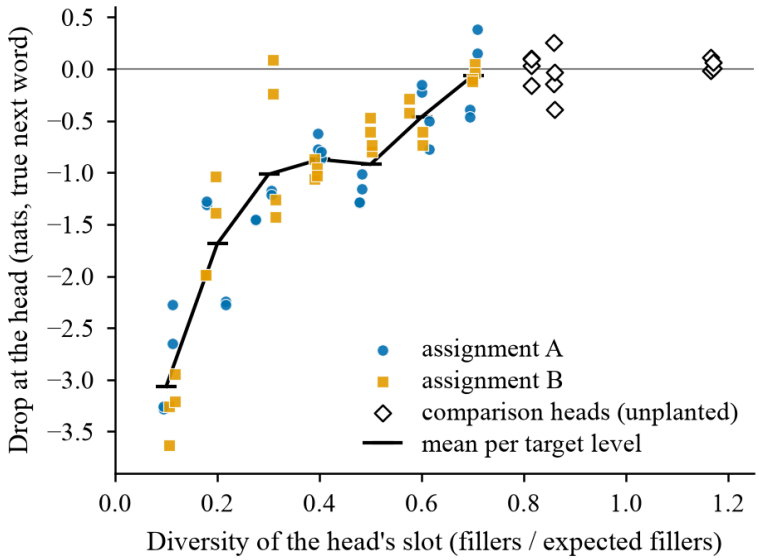


Figure 3. Dose and response on FineWeb. Fourteen heads planted to diversity 0.1–0.7 (two assignments, two seeds each), against the drop in the log probability the model gives the true next word at the head, relative to a model trained on the unplanted corpus. The loss is monotone in the dose (Spearman 0.76–0.92) and does not vanish at the mild end. Unplanted comparison heads mostly stay near zero.

5.3 Realistic cuts of a corpus

Five cuts of 10–50% of F were diagnosed against F, and six models were trained per seed (§4.2).

Cut	Tokens/word	Under-shown	Absent	Class rows flagged
random 50%	1.49	0	0	0
random 25%	1.49	0	0	0
random 10%	1.52	0	0	0
least Wikipedia-like 25%	1.39	2	11	4
most formulaic 25% (lowest perplexity)	1.61	1	4	1

Random cuts read as random. They flag nothing, neither rows nor classes.

Filters are the only cuts flagged, but mildly. The Wikipedia-like filter flags "," and ":" before determiners and labels, and the formulaic filter flags ". Our". Their absent rows are a few thin heads ("50", " a day", " they should"). We had predicted at least 10 under-shown rows for each filter, and both predictions failed.

The z of a head predicts what the cut costs the model better than its frequency. In the 10% cut, Spearman(z, drop) over 416 heads is 0.157 / 0.181 (two seeds), and that of log trunk alone is 0.037 / 0.018.

Filters move slots. A cut's score is its heads' excess E = D − mean D over 8 random cuts of F of its size, weighted by F's filler shares, and z divides it by the draws' leave-one-out spread. The perplexity filter that keeps the 25% most formulaic documents reads z = +18.6, the one that keeps the documents least like Wikipedia reads z = +24.8, and the CCNet-style Wikipedia filter reads z = +9.7. Random cuts read $|z| \leq 1.5$.

The model pays where the excess points. Within a filtered cut, heads with larger E lose more than a random cut of the same size costs: Spearman(E, drop − random drop) ranges from −0.24 to −0.37, on six filters and both seeds. Raw D's apparent success on random cuts was sample size alone. In the 10% cut, D reads −0.42 / −0.45 against the drop, and E only −0.05 / −0.11.

Choosing among cuts. Over ten 25% cuts, the corpus score orders the models' bits per byte only partly: Spearman 0.65 / 0.55, against a prediction of at least 0.5 on both seeds. Document count, a proxy for topical coverage, matters as much (−0.47 / −0.44), and the score orders what document count leaves at 0.58 / 0.56. Along one filter's dose, with documents held fixed, the score orders the models (Spearman 0.90); when the dose also adds documents, it does not (0.60).

5.4 Syntax is learned at the class level

First reading, negative. Among realistic cuts, G followed size (F 0.748, 25% cuts 0.700–0.736, 10% 0.69) and document count, not the score (−0.16 / −0.33). G per head reproduces across seeds (Pearson 0.87–0.91). The reason, which we found next, is that realistic filters thin a head's slot but never empty the pattern of its class.

Head versus class removal. From r50, at matched bytes, we build three corpora. The class cut removes every sentence where a pronoun (*I you he she we they*) or a modal is followed by a verb; the head cut removes only those of *she, they, can, will*; and the control removes random sentences.

Class	Class cut ΔG	Removed heads: head cut / class cut	Siblings in head cut
pronouns	−0.195	−0.044 / −0.177 (25%)	−0.028
modals (VB items)	−0.126	−0.040 / −0.081 (49%)	−0.008
possessives	−0.303	−0.194 / −0.343 (57%)	+0.046

Removing the class pattern removes the syntax of every member. Removing it from two heads costs them a quarter to a little over half as much, and leaves their siblings intact.

The class level of the indicator agrees with the models. In the class cut the tool reports the modal node (t 319) and the pronoun node *I you we they* (t 254), against a class bar of 13.9. In the head cut it reports no pronoun node, and a five-modal node holding *can* and *will* at t 33, a tenth of the class cut's. Head-level D, by contrast, says *she* and *they* lost the same in both cuts (+1.46 / +1.58; earlier reading, Appendix B).

Syntax is therefore learned at both levels, mostly at the class. How much a head's own evidence adds depends on the class: least for pronouns, most for possessives.

5.5 Attribution: corpus or model?

The flagged class cuts and their unflagged size-matched controls are trained in three configurations (small, medium, long), alongside F. The tool flags the class rows of each class cut (pronouns and modals t 254 and 319, possessives t 165), and no class row of any control.

Class	Best on F	Best on control (unflagged)	Best on flagged cut
pronouns	0.981	0.961	0.846
modals	0.999	0.988	0.870
possessives	0.872	0.859	0.600

On the flagged cuts, no configuration recovers the class (0.13–0.27 below F). More capacity makes it worse: pronouns go 0.85 → 0.75 → 0.68 and modals 0.87 → 0.80 → 0.75, as a larger model memorizes the pattern's absence more firmly. On the unflagged controls, the best control is within 0.02 of F. On a smaller unflagged cut (10%), changing the model moved G by up to 0.06; that is the engineer's lever.

At ten times the data. BigF, its class cut and its control were trained with the medium model for 30,000 steps.

Model	BLiMP	G pronouns	G modals
BigF	0.629	0.910	0.902
control	0.632	0.822	0.895
class cut	0.622	0.761	0.565

BLiMP rises from 0.57 to 0.63. For modals, the attribution result is stronger at scale: the cut leaves modals near chance. For pronouns, the gap shrinks to 0.06, and our predictions failed. At this data size almost every frequent past-tense verb follows each pronoun in the reference, which leaves 3–7 unseen items per pronoun, and the per-head readings become erratic. We cannot separate low test power from a real reduction. More data also brings more indirect evidence (noun subjects), as FiCT reports. Figure 4 collects the class-level results of §5.4 and §5.5.

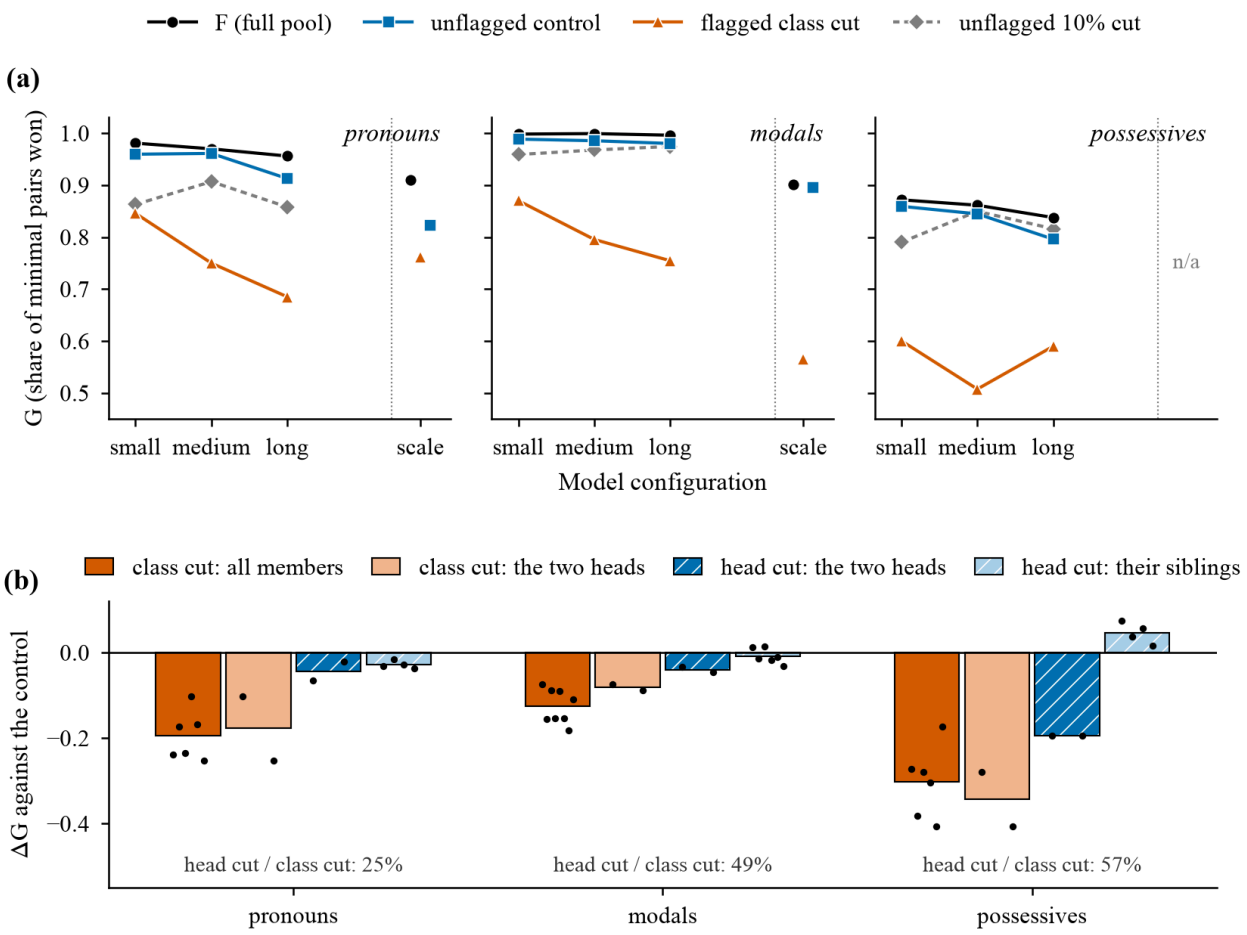


Figure 4. Corpus or model. (a) G of models trained on the full pool F, on unflagged size-matched controls, on the flagged class cuts and on an unflagged 10% cut, across model configurations; "scale" is the medium model trained on ten times the data (BigF and its cuts). No configuration recovers a flagged class, while unflagged controls stay within 0.02 of F. (b) Change in G against the control: removing a pattern from the whole class costs every member, while removing it from two heads costs them 25–57% as much and leaves their siblings intact.

5.6 The tool's detection

We diagnosed the 22 labeled cuts of §4.1 against F at 1.50 tokens per word. The class rows find every removed class, each in its own branch of the tree, and flag no control.

Cut	Node reported	t (class bar 13.9)
pronoun and modal class cut	*can will would may could should* (8 heads)	319
	I you we they i	254
possessive class cut	*your his my their*	165
number class cut	*two … seven*	104
base-verb class cut	*get make see take find leave*	101
pronoun and modal head cut	*can will would could should*	33
possessive head cut	*his their*, exactly the removed heads	85
number and base-verb head cuts	nothing	
12 random controls	nothing	

The head rows separate cuts from controls in the same way: the class cut and the head cut flag 39 and 13 head rows, and the 12 controls flag none. The *absent* rows find every one of the 8 dropped words.

On 8 fresh random controls, two rows are read absent, one in each of two controls: `luck`, shown 0 times against the draws' 7.0 ± 1.8, and `Maybe`.

Finally, we read 48 random whole-document cuts of F, which are random samples of the reference, against one fixed set of draws. They yield 2 head flags (4%), no class row and no *absent*. alpha is 0.06 per family.

5.7 Comparison with corpus metrics and with BLiMP

One-number metrics. On the class cuts, the head cuts and eight random controls, we computed perplexity under F's bigram and under a Wikipedia bigram, unigram KL from F, MATTR (Covington & McFall, 2010), distinct-2 (Li et al., 2016), and the Vendi score (Friedman & Dieng, 2023) of MiniLM sentence embeddings (Wang et al., 2020; Reimers & Gurevych, 2019). The planned test (3 sd of the controls) flags at most one class cut per metric. That test was lenient: the controls mixed sizes, and the mean perplexity had an outlier. In a size-matched reading, which we treat as exploratory, unigram KL and distinct-2 do register the two large class cuts, which removed 18,126 and 10,315 sentences (KL +0.019 and +0.011 against noise of about 0.001). They stay within noise, however, on the numbers (1,972 sentences) and the base verbs (3,642), which the indicator flags at t 104 and 101, and no metric says which pattern changed. The claim these results support is therefore localization and sensitivity to small, pattern-specific losses, not that other metrics see nothing.

BLiMP. Our small and medium models read 0.56–0.58 on BLiMP (chance 0.5), and 0.63 at scale; subject–verb agreement sits at about 0.52 at small scale. Between control and class cut, BLiMP moves at most 0.017, while G moves 0.11–0.34: a single score over 67 paradigms dilutes a class-specific loss. G and BLiMP also measure different syntax. G reads local slot classes, which our models learn well (0.96–1.00 for pronouns and modals), whereas BLiMP's paradigms mostly turn on longer dependencies.

5.8 No arbitrary settings

Each hand-set threshold of the earlier version (§4.4) was either derived from the data or measured against the verdicts on the 22 labeled cuts of §5.6.

Granularity. One tokenizer was learned on F to 60,000 merges and cut at each granularity.

Tokens/word	Merges	Classes found (of 5)	Controls with a class flag (of 12)	Controls with a head flag (of 12)	Dropped words
1.35	30,526	5	1	0	8/8
1.40	20,608	5	0	0	8/8
1.50	11,381	5	0	0	8/8
1.60	7,220	5	1	1	8/8
1.70	4,944	5	0	3	8/8
1.80	3,548	5	0	0	8/8

The verdicts hold over the whole range, and the control flags (6 of 144 readings) are near the designed rate. Between adjacent granularities, under-shown and within-noise swap in 26–37 of 330,000–690,000 patterns; what moves is whether a rare pattern is read at all. A granularity of 1.30 does not fit in 16 GB, as it leaves 45,748 tokens to cluster. On F alone, the range has no measured edge on either side; its edges come from the other domains, below. A fixed budget, by contrast, means nothing across corpora: 4,000 merges leave 1.76 tokens per word on F, while on 30 recipe documents

the merges run out at 269, near whole words. The tool therefore learns to a granularity, not a budget, and refuses a budget outside the range.

Domains, size and draw. We repeated the reading at 11 granularities from 1.30 to 2.50 on five corpora: TinyStories, FineWeb, Wikipedia, and two knowledge domains, PubMed abstracts and the facts of European Court of Human Rights cases. Each corpus was read at 0.9, 3.6, 14.4 and 57.6 MB, on three disjoint pools at each size up to 14.4 MB and one at 57.6 MB, each pool with its own 22 labeled cuts: 550 readings in all. A granularity holds when every removed class is found and every dropped word that is one token is read absent.

Size	Pools where 1.50 holds	Range common to all corpora
0.9 MB	1 of 15	none
3.6 MB	6 of 15	none
14.4 MB	12 of 15 (14 of 15[1])	1.35–1.70
57.6 MB	5 of 5	1.35–2.00

[1] Leaving out a dropped word that no granularity of the pool reads absent (see below).

From 14.4 MB, the verdicts hold from 1.35 to 1.70 in all five domains, on every pool but one once unresolvable dropped words are left out[1]; FineWeb's third pool fails on one dropped word (below). 1.30 fails in FineWeb's second pool and PubMed's merges run out before it, and 1.80 fails in two of legal's three pools. The upper edge moves with the draw and grows with size: at 57.6 MB, Wikipedia holds to 2.50 and legal to 2.20. False flags on the controls stay under twice the test's level in every one of the 50 corpora. The tool's range is therefore 1.35–1.70, and 1.50 stays inside it.

Below 14.4 MB the readings are erratic, and the reference is the small pool itself. Removed classes are missed in 86 of the 134 failing readings at 0.9 MB, most often the base verbs and the numbers. The class tree's settings are not the cause: the noise floor of its cut never binds, and a context matched by count to the 14.4 MB measurement recovered TinyStories' possessives but none of Wikipedia's or legal's classes. The tool therefore warns when its domain holds less than 14.4 MB.

The failures at 14.4 MB are single dropped words whose uses sit in few documents: *assessment* has 59% of its uses in one FineWeb document, *retrieved* 68% in three, *oil* 80% in one Wikipedia article. Against random whole-document draws, the absence of such a word looks like a draw that missed its document, at every granularity. The eight words chosen by hand for F are spread over 5% to 15% of the documents, and none failed.

The class settings. We forced the coarse cut from 8 to 64 classes, and the context from 250 to 4,000 tokens (51% to 87% of running tokens). At every value all five classes are found, no control's class row is flagged, head flags on controls stay at 0–2 of 12, and all dropped words are found. Class rows do not depend on the coarse cut at all: they are read on the whole tree, and the class bar is 13.92 throughout. A head row, in contrast, is a head followed by one filler class, so a different cut asks a different question of the same head; pattern-level agreement with the 12-class cut falls from 97% (8 classes) to 63% (64).

The bar and the draws. With 16 draws and the maximum as bar, 6 of 48 random cuts were flagged against one draw set, and none with fresh draws for each cut. The bar had been calibrated on average over draw sets, but the maximum of 16 draws is itself noisy (5.6 to 9.6 across sets). A pool of 200 draws matched the Beta law of §3.4 at 16–128 draws. With 101 draws and $k = 6$, 2 of 48 are flagged.

What these replaced. The diversity floor of 0.5 and the 3-sigma margin would have needed a recalibration for every new corpus. When every token became a filler, the class floor of 0.5 nats missed every removed class but one, and no floor separated the class shifts that cost the models their syntax from those that did not. The minimum trunk of 20, swept from 2 to 40, changed no verdict between 5 and 40 on F but moved both bars, and it would not carry to a corpus of another size.

5.9 Without a reference

Saturation alone does not flag a thinned slot. We thinned the pronoun → verb pattern to 5–50% of its sentences at equal bytes and read each corpus on its own. The slot's missing mass rises from 0.028 to about 0.05, but the median node of every corpus reads 0.40, so any tolerance that flagged the thinned slot would flag most of the corpus. Removed outright, the slot reads 0.028 again, as saturated as the control.

Comparing a head with its siblings does flag a thinned slot. Read with the sibling test of §3.5, *they* in the head cut of §5.4 shows 3 verbs where a random split of its parent gives 26.5 (t 9.6, bar 6.3). The tool flags *she* and *they*, and no pronoun of the control. The flags replicate: the control's two random halves, read on the control's own tree and classes so that every test is the same test, share 264 weak tests, of 332 and 397. On the recipe grammar, 11 of 14 restricted verbs are weak; the other three share their only ingredient with their sibling, and no verb is weak in the diverse recipes. On FineWeb the tool flags 95 head rows in the control and 120 in the head cut, in 2.5 minutes.

Most of these flags are heads that grammar narrows within their filler class: *to* takes fewer fillers than the other prepositions (524 against 871), and reporting verbs (*said, says, means*) fewer than other verbs. They are weak in the corpus, which is what a model trained on it sees. A reference reads them within noise, because the domain has them too.

The test cannot see some cases. Thinning every pronoun at once leaves no sibling stronger: the pronoun node reads 0.94–1.06 of its expected fillers when thinned to 25% or 10%, and the class cut shows its slot 19 times. In a synthetic subject–verb corpus, *she* and *they* restricted to two verbs form their own class and lose their sibling.

We rejected three alternatives. Comparing each head with its whole coarse class, adding a test of how often a head takes each filler class, and grouping heads by context flagged 500–2,000 heads and 3,000–15,000 class rows in the control, every pronoun among them. Their halves shared 12–17% of the flags.

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6. Discussion

What the indicator certifies. A flagged class pattern is a corpus fault. In our experiments no configuration recovered it, and capacity made it worse. The remedy is data: text in which the class's heads take other fillers of their slot. Conversely, an unflagged corpus lets the reference's syntax be reached at its size, and a failure there points at the model. This division of labor is the practical point of the work. The data side certifies that the patterns are served; finding the model that learns them is left to model engineering.

Why class level. Models share evidence across heads of a class: a head with a thin slot borrows from its siblings, which agrees with the indirect-evidence results of FiCT and the AANN study. Realistic filters thin heads but rarely empty classes, which explains why, among realistic cuts, class-level syntax followed size and topic rather than the score (§5.4). A diagnostic aimed at syntax must therefore read

classes, while a diagnostic of which word a model will predict reads heads. Reading every node of the head tree, rather than one cut, lets the report name the group that carries a shift: the whole modal class in one cut, two possessives in another.

Settings from the data. A threshold set on the corpora where a method was developed is a claim about those corpora. The indicator's bar instead says, for each corpus, how far a random sample of the reference strays at its worst. The reader is left with one number to judge, alpha, which is a tolerance and not a property of the data. The cost is time: 101 draws make a diagnosis 4–9 minutes at our sizes, against about 4 minutes before.

With and without a reference. Without a reference, the corpus is judged by its own evidence: a head weaker than its siblings is weak for a model trained on it. This is the reading a practitioner has before they find a reference, and it flags real gaps (*she* and *they* in the head cut). A reference does two further things. It sees what the corpus lacks altogether, or lacks for a whole class, and it separates the corpus's weakness from the domain's. Both readings are relative. "Unflagged" means as good as a random sample of the reference of this size, or as good as a random share of the head's parent; it does not mean as good as the reference at any size. In our data, a random cut of a large, well-built corpus already preserves the patterns as well as any filter. The indicator's practical role is mainly to reject cuts and filters that bias syntax.

Effect sizes. Whole-model effects of slot damage are small in bits per byte (0.017 over a whole filter ladder), whereas per-pattern effects are large: 1–4.6 nats for planted heads and 0.13–0.33 in G for class removals. A per-pattern diagnostic is needed precisely because whole-corpus and whole-model numbers dilute local losses.

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7. Limitations

The first limitation is scale. The models go up to 6.2M parameters, and the corpora up to about 10M words, on CPU, and on BLiMP the models read 0.56–0.63. Our claims therefore concern local syntax learned by small models; whether the class-level attribution holds for models that learn long-range syntax is untested. The indicator itself covers local syntax only, that is, heads of one or two tokens and the next filler. Long-range dependencies (agreement across attractors, filler–gap, binding) are out of scope.

The model-side evidence comes from one language and one natural source, English and FineWeb, plus a synthetic grammar. On the corpus side, the granularity range was measured in five English domains, from 0.9 to 57.6 MB, with three draws per size up to 14.4 MB and one at 57.6 MB; other languages and larger corpora are untested. Below 14.4 MB the verdicts do not hold, so the tool needs a domain of at least that size. A dropped word whose uses sit in a few documents cannot be read absent at any granularity: the indicator resolves the absence of words spread over many documents, not of words concentrated in a few. Only some classes were tested at depth: three (pronouns, modals, possessives) were trained under removal, and two more (numbers, base verbs) were checked on the corpus side. At scale, the pronoun result was not confirmed, because its test lost power. Most experiments used two seeds; the new model configurations and the scale runs used one.

The test's level, alpha = 0.06, is per family. A random corpus can be flagged in any of three families, so its chance of some flag is at most 18%; measured, it was 4% (2 of 48). Splitting alpha three ways would need about 300 draws. In the planted slices, one class row was flagged in each of two of seven unplanted slices. The granularity range's upper edge moves with the draw and grows with size (to

2.00 at 57.6 MB), so 1.70 is the edge common to every domain at 14.4 MB, not a property of each corpus.

Without a reference, a pattern the corpus lacks, a class shown weakly as a whole, and a head the tree sets apart from its siblings are not seen. As for D on plantings, the dose fit of §5.2 (drop $\approx -0.68 \cdot D$) and the head-level D of §5.4 come from the earlier version's reading of word-like tokens (Appendix B) and were not rerun with the final tool.

Two further limits concern what the indicator measures. On the recipe grammar, the diversity a model needs grew with model size. The indicator says where a corpus departs from its reference; how much departure a given model tolerates is not part of it. Finally, the indicator addresses syntax, not semantics. A corpus can pass and still fail a model on topic or meaning: a topic-narrow corpus failed models document-wide without any slot flagged.

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8. Conclusion

We asked whether a corpus can be checked, before training, for the local syntactic patterns its domain needs, so that a generalization failure can be attributed to the corpus or to the model. The answer, within the limits above, is yes.

An indicator that compares each slot's fillers with random samples of a reference, at every node of an induced tree of heads, flags corpus faults that no model configuration we tried recovers. Where it flags nothing, a model reaches the reference's syntax. None of its settings is set by hand: the granularity, the trunk read and the number of samples come from the data, and the test's level is stated in every report. Without a reference, the same tree lets a corpus be judged by its own evidence. Along the way, we found that models learn local syntax mostly at the class level, which explains both why realistic data filters rarely break syntax and why a diagnostic must read classes. The tool, code and every preregistered prediction, including the failed ones, are released with the paper.

Future work includes larger models and corpora, with GPU; long-range dependencies; more languages, to test that the range measured in five English domains carries; the tolerance of a given model size to a flagged departure; and a semantic counterpart for knowledge domains.

Appendix A. Predictions that failed

The predictions are kept as written in the preregistration documents.

Indicator and models (earlier version of the indicator).

- **Recipe grammar.** The minimal pairs "cut the slowly" did not stay near 100% (0.25 with one filler). The corpus-size predictions failed at 500 documents.
- **Plantings.** One comparison head moved -0.51 nats. Whole-text loss rose 0.03 bits per byte (predicted < 0.02). Detection at $k = 32$ was still frequent (12/20).
- **Topic-narrow corpus.** The flagged heads did not lose more; the loss was topical and document-wide.
- **Subset selection.** Mild natural flags (diversity 0.26–0.50) did not cost the model. In r10, Spearman(z, drop) was 0.10 / 0.12 (bar 0.15).

- **Dose.** The mild end cost 0.38–0.57 nats (predicted ≤ 0.3), and the threshold was crossed at 0.64.
- **Divergence.** On plantings, D ordered heads no better than diversity. Its first cut, κ, failed because of a sample-size floor.
- **Cut choice.** Over ten cuts, Spearman(score, bits per byte) was 0.55 / 0.45 (predicted ≥ 0.5 on both seeds). With the current reading it holds (0.65 / 0.55).
- **Ladders.** The ladder that adds documents had Spearman 0.60. The per-head reading failed at dose 25%.
- **Syntactic evaluation of realistic cuts.** All six predictions failed, which led to the class-level hypothesis.
- **Class patterns.** P2 came out at −0.086 (bar −0.10), with an invalid first modal test. In P4, the noise between models exceeded the bar.
- **Replication.** The possessive head cut cost 57% (bar 50%).
- **Tool.** The pronoun class was missed (+0.47 against a 0.5 floor). The earlier *absent* rule gave 79–114 false flags, and one control gave 5 at m/2.
- **BLiMP.** F's models read below 0.60.
- **Scale.** The pronoun gap was 0.06 (bar 0.10), and the control was 0.09 below the reference.

Settings from the data.

- **Granularity.** Adjacent granularities were predicted to agree on 90% of patterns; they agreed on 85–94%, and almost all of the difference was whether a pattern is read. With a trunk of 20, the numbers and base verbs were missed at 1.35, although they were predicted to be found over the whole sweep.
- **Class floor.** Each class whose removal cost the models their syntax was predicted to read a larger excess than every one that did not. The prediction failed: at 64 head classes over every token, pronouns and possessives shared one class with *the* and *a*, and moved it by +0.045 and +0.007. This led to class rows at every node.
- **The 16-draw bar.** 6 of 48 random cuts were flagged against one draw set (design: about 1 in 17). This led to alpha and 101 draws.
- **Without a reference.** Saturation (missing mass) did not flag a thinned slot, and three groupings of heads gave flags that the corpus's halves did not share.
- **Subset selection, rerun.** Both filters were predicted to flag at least 10 rows; they flagged 2 and 1.
- **Domains, size and draw.** 1.50 was predicted to hold at every size and on every draw; it held on 1 of 15 pools at 0.9 MB and 6 of 15 at 3.6 MB, and failed on three of fifteen 14.4 MB pools, each on one dropped word concentrated in a few documents. The 57.6 MB readings were predicted to match 14.4 MB; Wikipedia's and legal's ranges widened instead. PubMed was predicted to need more merges than Wikipedia to reach 1.50 (15,066 against 16,022), and its range and legal's were predicted to contain 1.30–2.00 (legal's ends at 1.70). This led to the 14.4 MB minimum and the range 1.35–1.70.

Appendix B. Earlier version of the indicator

The indicator was first built with hand-set thresholds (§4.4). Its results, replaced in the body by those of the final tool, were the following.

- **Planted gaps (§5.2).** With a 3-sigma margin, the replacement sweep found 5, 5, 5, 5 and 4 of 5 heads on FineWeb at k = 2, 4, 8, 16 and 32, and 19, 20, 19, 17 and 12 of 20 heads over four sources (FineWeb, Wikipedia, arXiv, C4). Removing sentences found 3 of 5: two heads left with 6 and 23 occurrences read just above a fixed diversity floor of 0.5.
- **Realistic cuts (§5.3).** The 3-sigma margin flagged 2–8 rows on each random cut, at the rate chance gave it. In the 10% cut, Spearman(z, drop) was 0.10 / 0.12, with log trunk alone at 0.10 / 0.11. Reading word-like tokens only, the cut scores were $z = +15.0$ (formulaic filter), $+20.6$ (least Wikipedia-like) and $+3.9$ (CCNet-style), and $|z| < 0.6$ for random cuts. Spearman(E, drop − random drop) was -0.21 to -0.38, and Spearman(score, bits per byte) over ten cuts was 0.55 / 0.45.
- **Tool detection (§5.6).** With 64 head classes and a 0.5-nat class floor, the class rows found 4 of 5 removed classes, missing the pronouns at $+0.47$ nats. The possessives were found only through their short heads, because they shared an induced class with *the* and *a*. The first *absent* rule fired 79–114 times on random controls.
- **Still in the body.** Two numbers come from this version's reading of word-like tokens and were not rerun: the dose fit of D (drop $\approx -0.68 \cdot D$, R^2 0.77; §3.3, §5.2) and the head-level D of §5.4.

Appendix C. Reproducibility

Every experiment has its own script, its results (JSON) and a record of its design, predictions and outcome, written before it ran. The diagnosis is packaged as a command-line tool (`corpus-diagnose`) with a web interface. Runs are CPU-only; a diagnosis takes 4–9 minutes at our sizes after reading the reference once. Code, data and preregistration records are available from the author on request and will be released with publication.

Declaration of generative AI use

During this work the author used Claude (Anthropic) to assist in writing code, running and analyzing experiments, and drafting the manuscript. The author directed the study, reviewed and edited all content, and takes full responsibility for it.

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